

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

ROTATIONAL MOTION

LECTURE-6

PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

- 1** Revision
- 2** Gabbar Numerical Series PYQs
- 3** ..
- 4** ..
- 5** ..

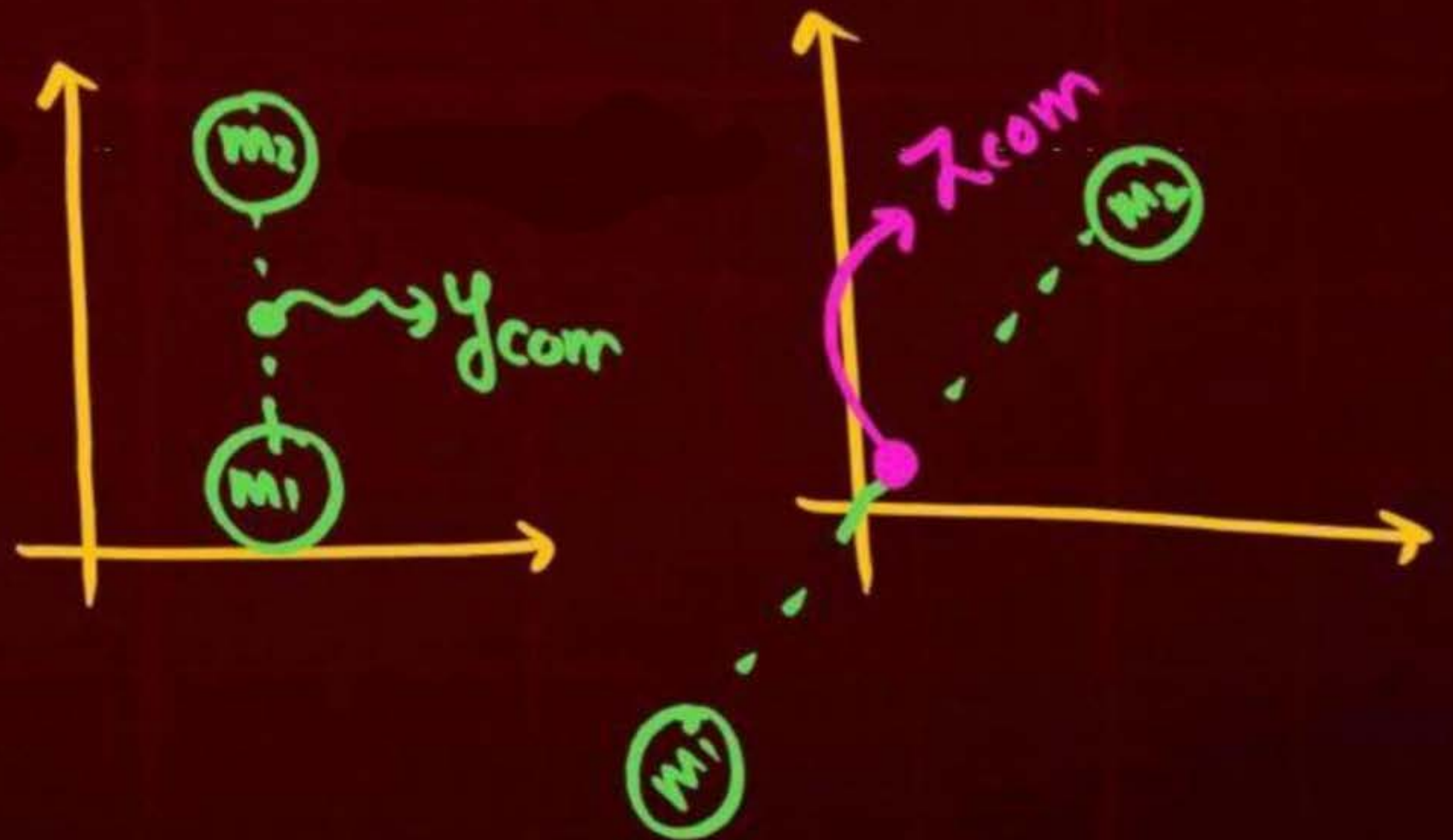
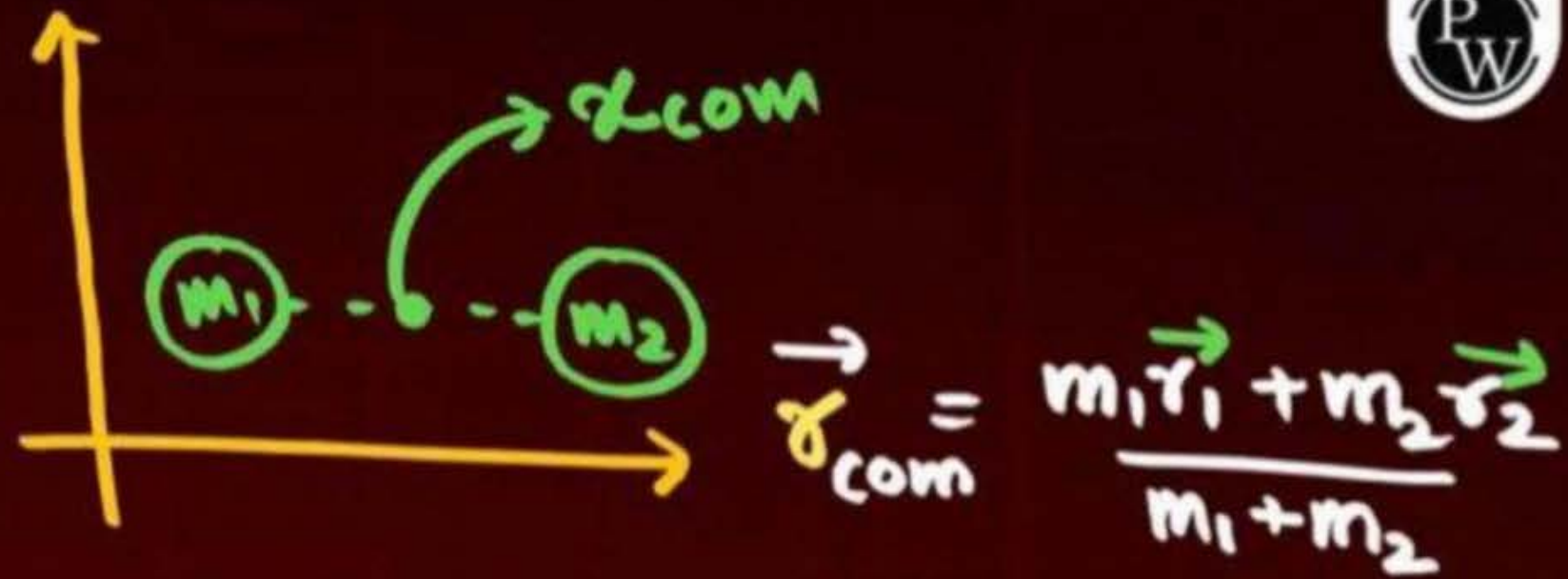
Recap From 1st Lecture

Position of Com

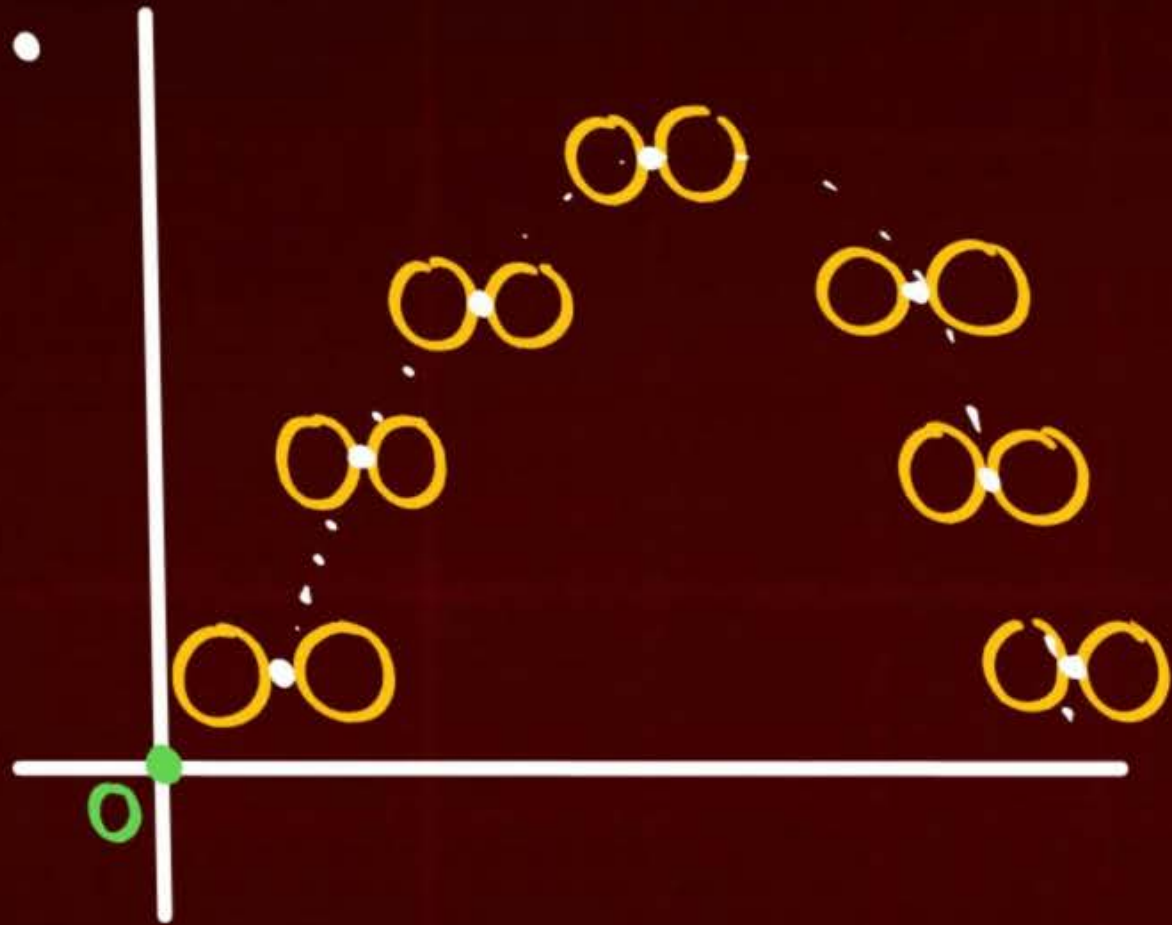
$$x_{com} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$y_{com} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

$$z_{com} = \frac{m_1 z_1 + m_2 z_2}{m_1 + m_2}$$



RECAP FROM 2nd LECTURE



① Formula $V_{com} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$

② $a_{com} = \frac{m_1 a_1 + m_2 a_2}{m_1 + m_2}$

Rotational Motion



✓ Angular displacement

$\theta = \frac{\text{Arc}}{R}$ $x = R\theta$ Rad

✓ Angular velocity

$\omega = \frac{\theta}{t}$ $v = R\omega$ Rad/sec

✓ Angular acceleration

$\alpha = \frac{\omega}{t}$ $a = R\alpha$ Rad/sec²

Recap From 3rd Lecture

Angular momentum

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\vec{L} = r p \sin \theta \hat{n}$$

Screw Rule



Torque

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\tau} = r f \sin \theta \hat{n}$$

numerical

$$\checkmark \vec{r} = 2\hat{i} + 3\hat{j} \quad \vec{p} = 3\hat{i} + 4\hat{j}$$

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\checkmark \vec{r} = 2\hat{i} + 3\hat{j} + 5\hat{k} \quad \vec{p} = 3\hat{i} + 4\hat{j} + 3\hat{k}$$

$$\vec{L} = \vec{r} \times \vec{p}$$

Trick

Recap From 4th Lecture

Torque τ

$$\# \vec{\tau} = \vec{r} \times \vec{F}$$

$$\tau = r F \sin \theta \hat{n}$$

(numerical)

Relationship b/w $\vec{\tau}$ and $\vec{L}$

$$\vec{\tau} = \frac{d\vec{L}}{dt}$$

Equilibrium

$$F_{\text{net}} = 0 \text{ (Translational equilibrium)}$$

$$\tau_{\text{net}} = 0 \text{ (Rotational equilibrium)}$$

Couple

Moment of Inertia

$$I = mR^2$$

$$\text{Ring} = mR^2$$

$$\text{Hollow sphere} = \frac{2}{3}mR^2$$

$$\text{cylinder} = mR^2$$

$$\text{Solid sphere} = \frac{2}{5}mR^2$$

$$\text{disc} = \frac{mR^2}{2}$$

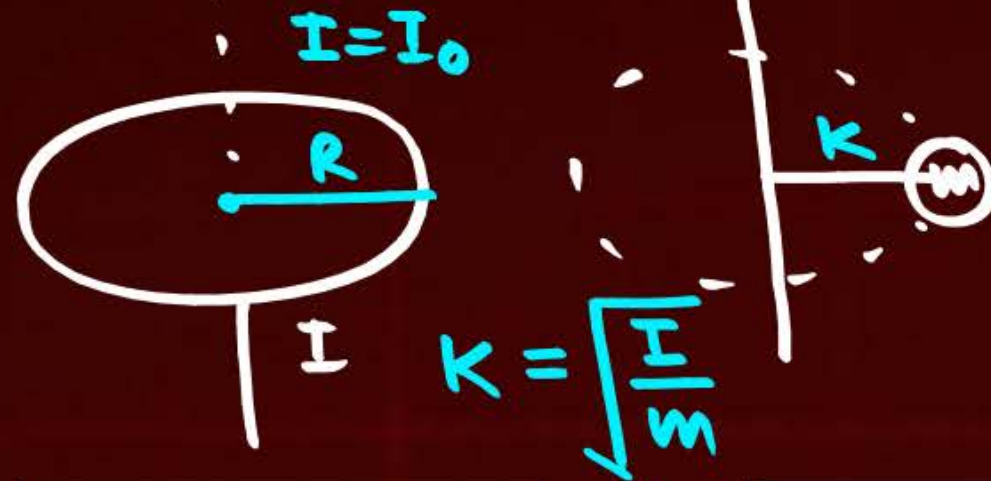
$$\text{Solid cylinder} = \frac{mR^2}{2}$$

$$\text{Rod (middle)} = \frac{1}{12}ml^2$$

$$\text{Rod (End)} = \frac{1}{3}ml^2$$

Recap From 5th Lecture

Radius of Gyration



Translation

$$x, v, a, m, F, p, W$$

$$K \cdot E = \frac{1}{2} m v^2 \checkmark$$

Rotation

$$\theta, \omega, \alpha, I = m R^2, \tau, L, W = \tau \theta$$

$$K \cdot E = \frac{1}{2} I \omega^2 \checkmark$$

$$\textcircled{1} v = u + at \quad \omega = \omega_0 + \alpha t$$

$$\textcircled{2} s = ut + \frac{1}{2} at^2 \quad \theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\textcircled{3} v^2 = u^2 + 2as \quad \omega^2 = \omega_0^2 + 2\alpha\theta$$

Law of Conservation $\vec{L}$

$$\tau = \frac{dL}{dt}$$

$$0 = \frac{dL}{dt}$$

$$0 \times dt = dL$$

$$0 = dL$$

$$0 = \Delta L$$

$$\Delta L = 0$$

$$L_f - L_i = 0$$

$$L_f = L_i$$



Gabbbar

Numerical Series

PYQs

QUESTION [SUMMER ZONE SET-A 2026]

Radius of gyration is denoted by:

A R

B G

C K

D I

$$K = \sqrt{\frac{I}{m}}$$

QUESTION [SUMMER ZONE SET-A 2026]

Explain the concept of torque and angular momentum.

QUESTION [SUMMER ZONE SET-A 2026]

✓ Briefly explain equilibrium of a rigid body. – 3 mark

QUESTION [SUMMER ZONE SET-B 2026]

A body in rotational motion possesses rotational kinetic energy given by:

- A** $KE = \frac{1}{2} I \omega^2$ ✓
- B** $KE = \frac{1}{2} I^2 \omega^2$
- C** $KE = 2 I^2 \omega$
- D** $KE = \frac{1}{2} I^2 \omega$

QUESTION [SUMMER ZONE SET-C 2026]

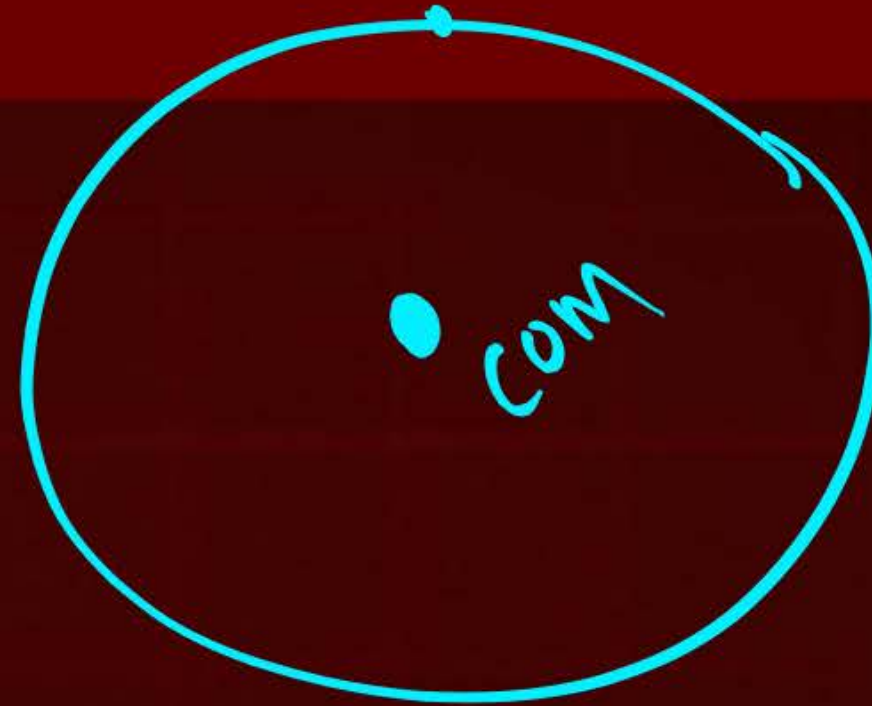
Explain the concept of torque and angular momentum.



QUESTION [SUMMER ZONE SET-C 2026]

The centre of mass of a bangle

- A** Lies inside the body
- B** Lies outside the body always
- C** Lies on the surface of body always
- D** None of the above



QUESTION [WINTER ZONE SET-A 2025]



Define angular momentum. Give its unit and dimensional formula.

QUESTION [WINTER ZONE SET-A 2025]

Write the comparison of translational and rotational motion.

QUESTION [WINTER ZONE SET-A 2025]

State the physical significance of moment of inertia.

Moment of Inertia is the resistance
to Rotational Motion

$$\# I = mR^2$$

$$\text{kgm}^2$$

$$ML^2$$

$I \uparrow$
Resistance

Rotation $\downarrow \downarrow$

$I \downarrow$
Resistance $\downarrow$

Rotation $\uparrow \uparrow \uparrow$

QUESTION [WINTER ZONE SET-C 2025]



State and establish the conservation law of angular momentum.

QUESTION [WINTER ZONE SET-C 2025]

Write the equation for momentum, and work done in rotational motion.

Rotational

$$\text{Angular momentum} = \vec{r} \times \vec{p}$$

$$\text{Work} = \tau \theta$$

QUESTION [WINTER ZONE SET-C 2025]



State the differences between Mass and Moment of inertia.

Ans

Mass

Total Quantity in Body

unit = Kg

Dimension = $M^1 L^0 T^0$

Moment of Inertia

Resistance to Rotational motion

⇒ $I \uparrow R \downarrow$ $I \downarrow R \uparrow$

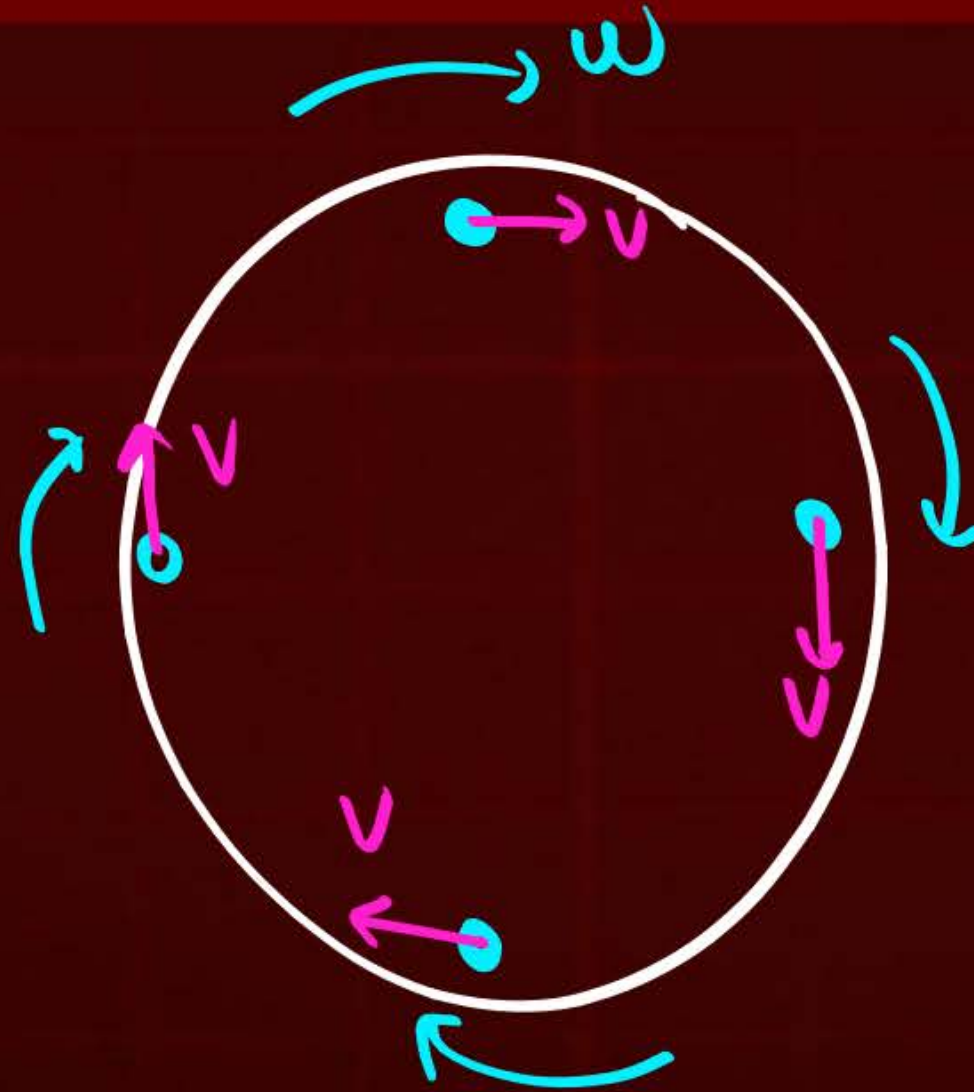
$I = mR^2 = \text{Kg m}^2$

Dimension = $M L^2$

QUESTION [JAMMU SET-A 2025]

Linear velocities of all the particles of the rigid body in rotational motion is:

- A** One ☒
- B** Zero ☒
- C** Same ☒
- D** Different ☒



QUESTION [JAMMU SET-A 2025]



A torque of 20 N-m is applied on a wheel initially at rest. Calculate angular momentum of the wheel after 3 seconds.

Sol

$$\tau = 20 \text{ Nm}$$
$$\omega_0 = 0$$
$$L_f = ?$$
$$t = \underline{3 \text{ sec}}$$
$$L_i = I_i \omega_i$$
$$= I_i \omega_0$$
$$= I_i \times 0$$
$$= 0$$

$$\tau = \frac{dL}{dt}$$
$$\tau = \frac{\Delta L}{t}$$
$$\tau = \frac{L_f - \cancel{L_i}^0}{t}$$
$$20 = \frac{L_f}{3}$$

$$20 = \frac{L_f}{3}$$

$$3 \times 20 = L_f$$

$$\underline{\underline{60}} = L_f$$

$$L_f = \underline{60} \frac{\text{kg m}^2}{\text{sec}}$$

QUESTION [JAMMU SET-A 2025]

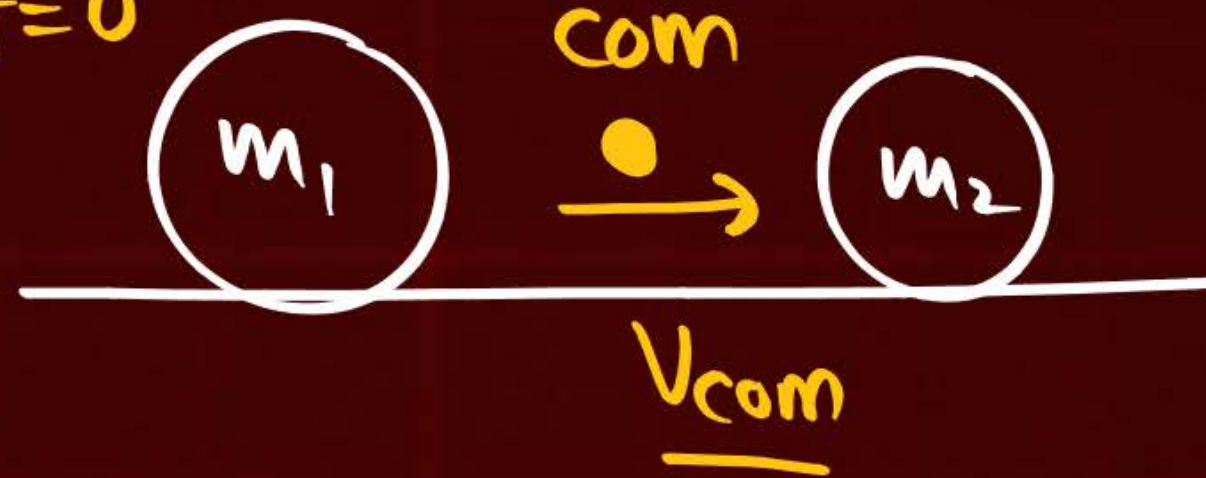
State and explain the principle of conservation of angular momentum. ✓

QUESTION [JAMMU SET-B 2025]

Centre of mass of an isolated system has a:

$$F=0$$

$$\tau_{\text{net}}=0$$



- A** Increasing velocity
- B** Decreasing velocity
- C** Constant velocity
- D** Variable velocity

QUESTION [JAMMU SET-C 2025]

A body in rotational motion possesses rotational kinetic energy given by:

A ☒ $\frac{1}{2} I \omega^2$

B $\frac{1}{2} I^2 \omega$

C $2 I^2 \omega$

D $I \omega$

$$K.E = \frac{1}{2} I \omega^2$$

QUESTION [JAMMU SET-A 2024]



Angular momentum is a:

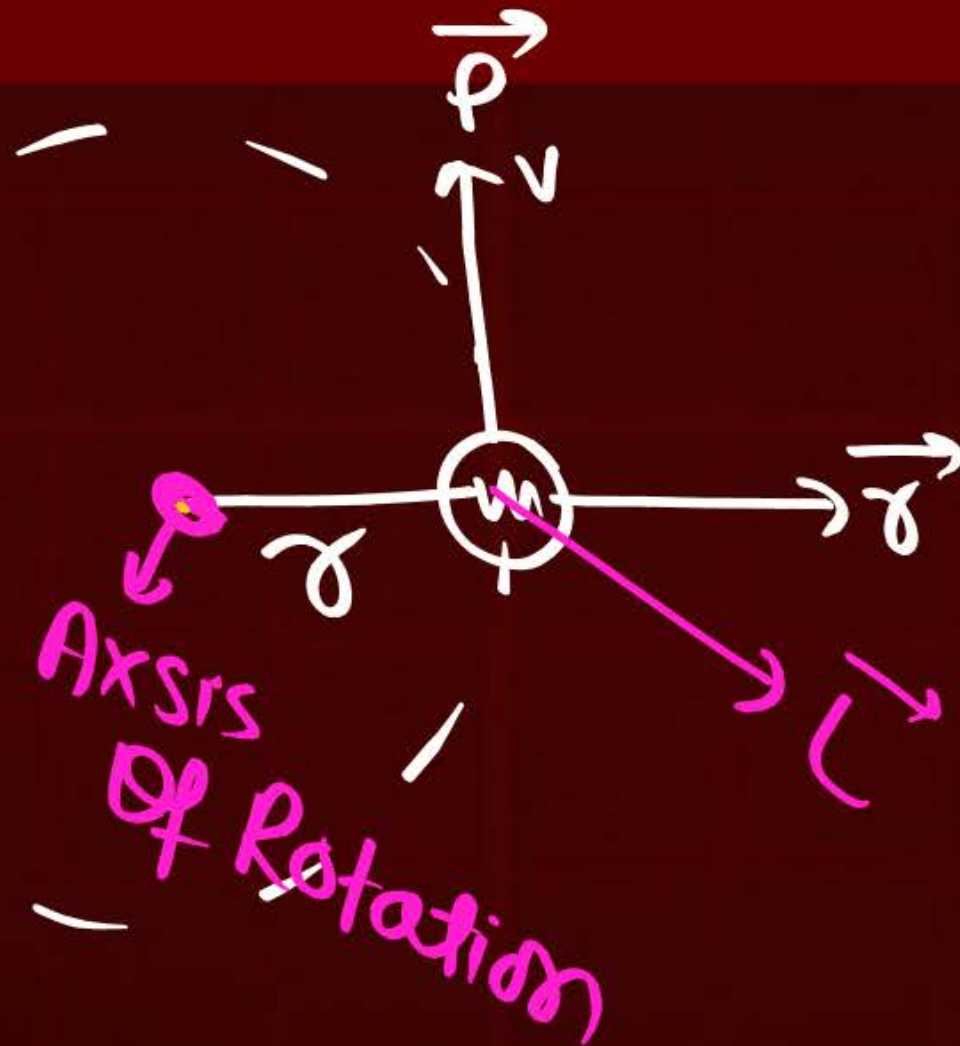


A Polar vector

B Axial vector

C Scalar

D None of these



$$\vec{L} = \vec{r} \times \vec{p}$$

$$\vec{L} = r p \sin \theta \hat{n}$$

Screw Rule

The Direction $\vec{L}$ is along the Axis of Rotation

Axial vector

QUESTION [JAMMU SET-A 2024]

Two bodies of mass 10 kg and 2 kg are moving with velocities $(2\hat{i} - 7\hat{j} + 3\hat{k})$ m/s and $(-10\hat{i} + 35\hat{j} - 3\hat{k})$ m/s respectively. Find velocity of centre of mass.

Sol

$$m_1 = 10 \text{ kg}$$

$$m_2 = 2 \text{ kg}$$

$$v_1 = 2\hat{i} - 7\hat{j} + 3\hat{k}$$

$$v_2 = -10\hat{i} + 35\hat{j} - 3\hat{k}$$

$$V_{\text{com}} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

$$V_{\text{com}} = \frac{10(2\hat{i} - 7\hat{j} + 3\hat{k}) + 2(-10\hat{i} + 35\hat{j} - 3\hat{k})}{10 + 2}$$

$$= \frac{\cancel{20\hat{i}} - \cancel{70\hat{j}} + 30\hat{k} - \cancel{20\hat{i}} + \cancel{70\hat{j}} - 6\hat{k}}{12}$$

$$= \frac{30\hat{k} - 6\hat{k}}{12} = \frac{\cancel{24\hat{k}}}{12} = \underline{\underline{2\hat{k}}}$$

Concept



formula



+ - x ÷



answer

$\int y dx$

$\frac{dy}{dx}$

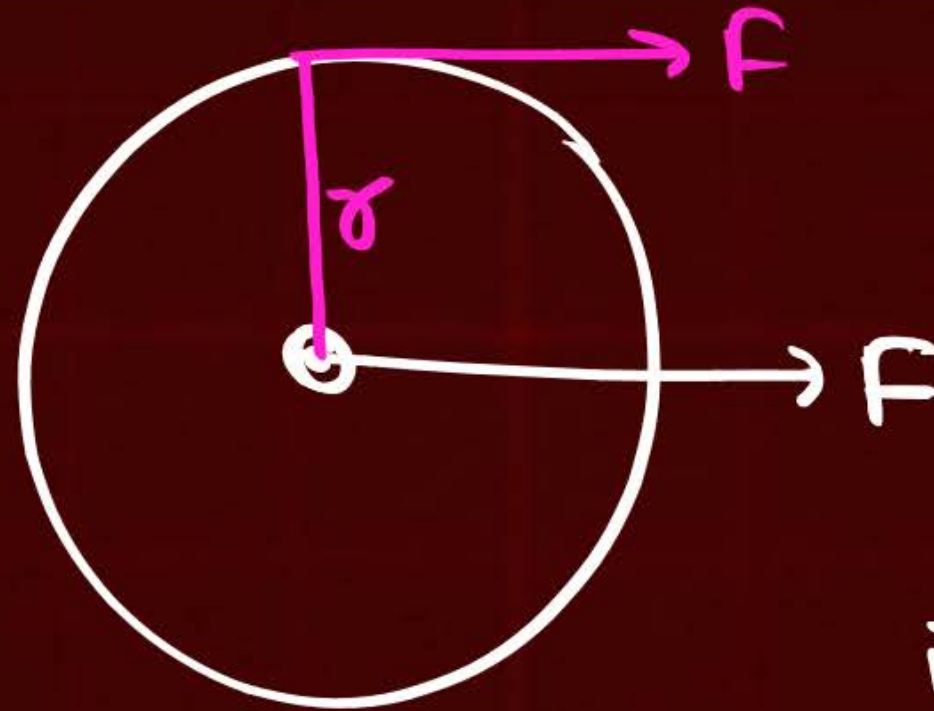
QUESTION [JAMMU SET-A 2024]

✓ Define centre of the mass. Find an expression for the position vector of centre of mass of a two particle system.

$$\vec{r}_{\text{com}} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

QUESTION [JAMMU SET-Y 2024]

Angular momentum of a particle rotating with central force is constant due to:



$$L = \text{constant}$$

$F \rightarrow \text{central}$

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\tau} = 0 \times \vec{F} = 0$$

$$\left. \begin{array}{l} \vec{\tau} = \frac{dL}{dt} \\ 0 = \frac{dL}{dt} \end{array} \right\} \begin{array}{l} 0 \times dt = dL \\ 0 = dL \\ 0 = \Delta L \end{array}$$

A

Constant force

B

Constant torque

C

Zero torque

D

Constant linear momentum

$$\underline{\Delta L = 0}$$

$$L_f - L_i = 0$$

$$\underline{\underline{L_f = L_i}}$$

$$\underline{\underline{L \rightarrow \text{const}}}$$

QUESTION [JAMMU SET-Y 2024]

The rotational kinetic energy of a body is given by $\frac{1}{2}I\omega^2$. Use this equation to obtain the dimensions of I .

Sol $\underline{K.E} = \frac{1}{2} \underline{I} \omega^2$

$\Downarrow$

$ML^2T^{-2} = I(T^{-1})^2$

$ML^2T^{-2} = I T^{-2}$

$\frac{ML^2T^{-2}}{T^{-2}} = I$

$I = ML^2$

$\omega = \frac{\theta}{t}$

$\omega = \frac{1}{T}$

$\omega = T^{-1}$

$I = mR^2$

QUESTION [JAMMU SET-Y 2024]

✓ (Neet)



Two bodies of mass 200 g and 500 g possess velocities $(20\hat{i} + 2\hat{j})$ and $(3\hat{i} + 5\hat{j})$ m/s respectively. Find velocity of centre of mass.

$$m_1 = 200 \text{ gm} = \frac{200}{1000} \text{ kg} = 0.2 \text{ Kg}$$

$$m_2 = 500 \text{ gm} = \frac{500}{1000} \text{ gm} = 0.5 \text{ Kg}$$

$$V_1 = 20\hat{i} + 2\hat{j}$$

$$V_2 = 3\hat{i} + 5\hat{j}$$

$$V_{\text{com}} = \frac{m_1 V_1 + m_2 V_2}{m_1 + m_2}$$

$$= \frac{0.2(20\hat{i} + 2\hat{j}) + 0.5(3\hat{i} + 5\hat{j})}{0.2 + 0.5}$$

$$= \frac{4\hat{i} + 0.4\hat{j} + 1.5\hat{i} + 2.5\hat{j}}{0.7}$$

$$= \frac{4\hat{i} + 1.5\hat{i} + 0.4\hat{j} + 2.5\hat{j}}{0.7}$$

$$= \frac{5.5\hat{i} + 2.9\hat{j}}{0.7}$$

$$V_{com} = \frac{5.5\hat{i} + 2.9\hat{j}}{(0.7)}$$

$x + y$

$$V_{com} = \frac{\cancel{5.5}\hat{i}}{\cancel{0.7}} + \frac{2.9}{0.7}\hat{j}$$

$$V_{com} = 7.86\hat{i} + 4.14\hat{j}$$

QUESTION [JAMMU SET-Y 2024]

The angular momentum of a moving body remains constant, if:

- ☐ A Net external force is applied
- ☒ B $\tau_{\text{ext}} = 0$
Net external torque is not applied
- ☐ C Net pressure is applied
- ☐ D Net external torque is applied

QUESTION [JAMMU SET-Y 2024]

(Neet = 2023)



Two spheres have identical masses and moment of inertia. One of them is solid and the other is hollow. Find the ratio of the radius of solid sphere to that of hollow sphere. $\frac{R_s}{R_H} = ?$



$$I_s = \frac{2}{5} m R_s^2$$

$$I_s = I_H$$

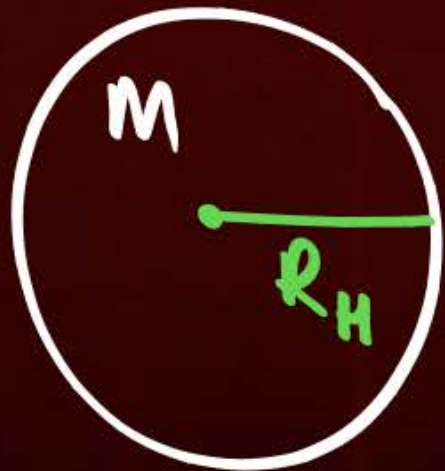
$$\cancel{\frac{2}{5}} m \cancel{R_s^2} = \cancel{\frac{2}{3}} m \cancel{R_H^2}$$

$$\frac{R_s^2}{5} \Rightarrow \frac{R_H^2}{3}$$

$$\frac{R_s^2}{R_H^2} = \frac{5}{3}$$

$$\frac{R_s}{R_H} = \frac{\sqrt{5}}{\sqrt{3}}$$

$$\underline{\underline{R_s : R_H = \sqrt{5} : \sqrt{3}}}$$



$$I_H = \frac{2}{3} m R_H^2$$

SUMMARY



Syllabus change

PYQ

PYQ $\Rightarrow$ PDF

HOMEWORK



$V_{com} \Rightarrow 2 \text{ numerical}$
 C_p

Love people

Use things not people

Don't use people and ^{↑ donot} love things

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